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**OCC SOFTWARE ENGINEERING Y2**

**SDT621: PM04**

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# Declaration Of Authenticity

I Fernando Laas (FULL NAME)

hereby declare that the contents of this assessment SDT621 PM04

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.

I have read and understood the Examination Rules and Regulations.

Signature: 

Date: 15/05/2026

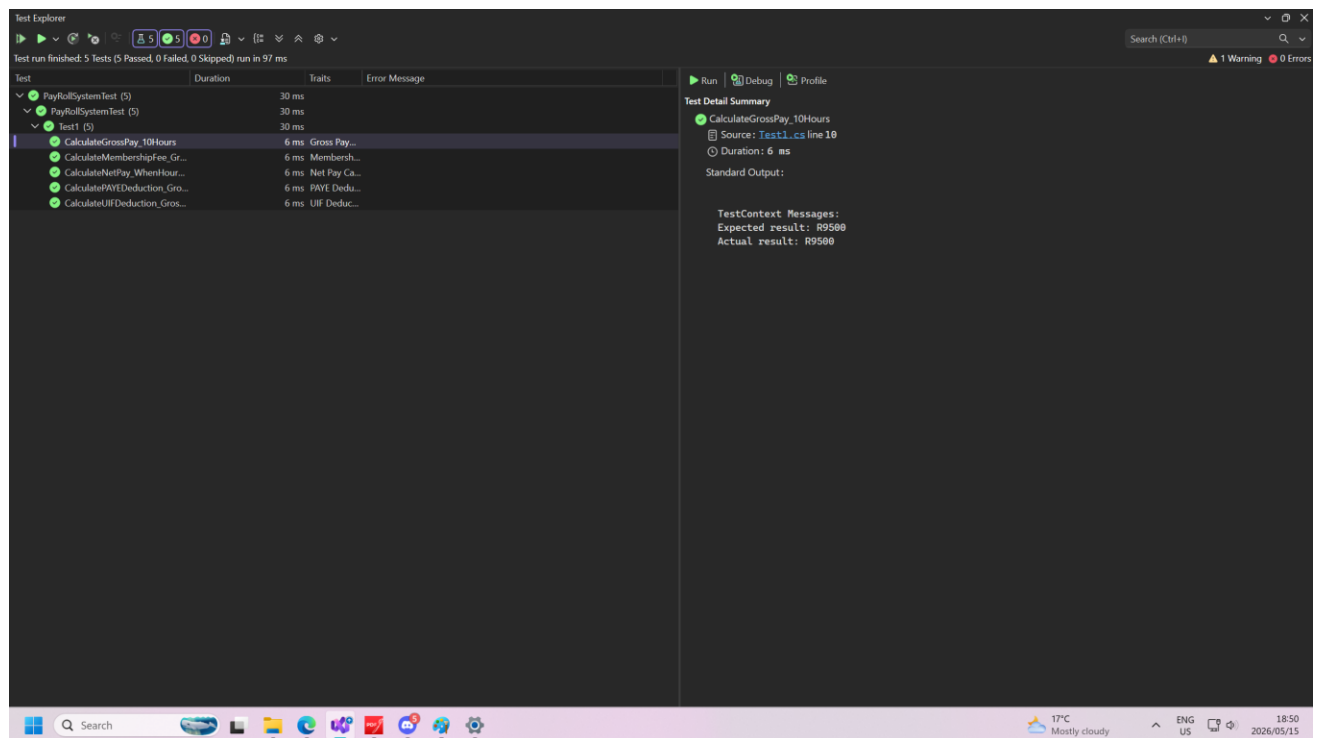
## GitHub Repo Link

[nandos622/PayRollSystem](https://github.com/nandos622/PayRollSystem)

# Unit Testing

| Test Case  | Input                               | Expected Output | Result   |
|------------|-------------------------------------|-----------------|----------|
| Gross Pay  | Hours worked = 10                   | 9500            | 9500     |
| UIF        | Gross pay = 10000                   | 100             | 100      |
| Membership | Gross pay = 10000                   | 1300            | 1300     |
| PAYE       | Gross pay = 10000<br>Dependents = 2 | 2212.5          | 2212.5   |
| Net Pay    | Hours worked = 10<br>Dependents = 2 | 6068.125        | 6068.125 |

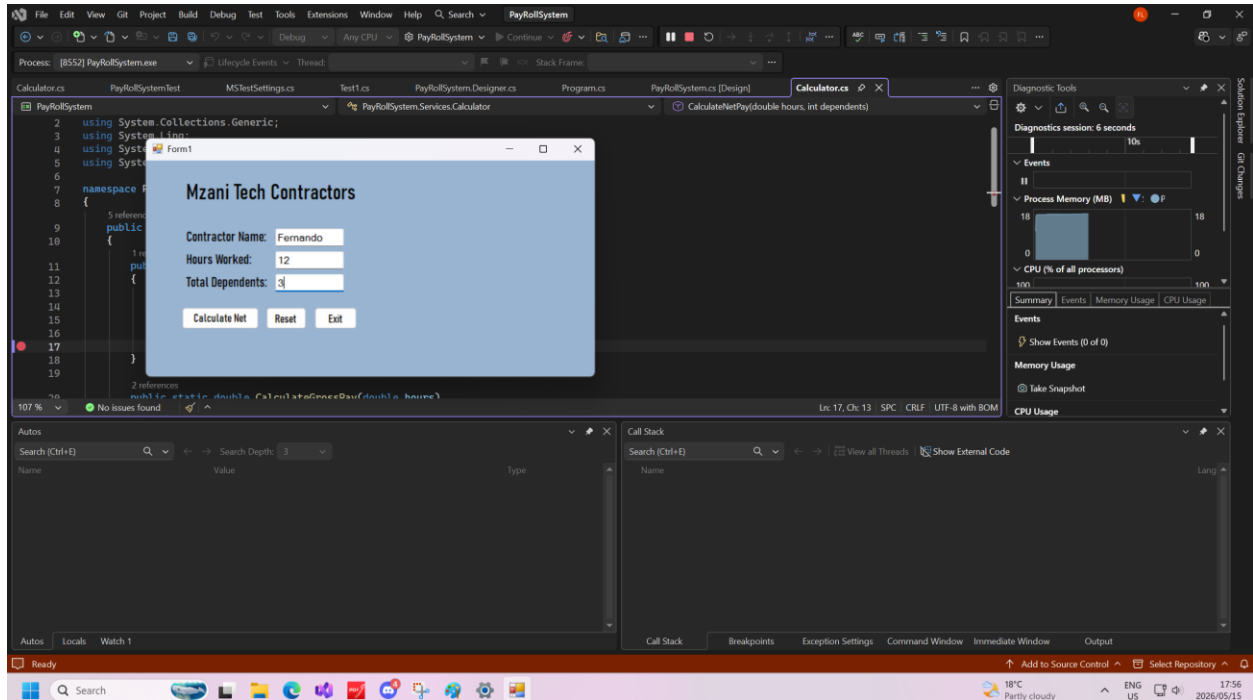
## Test Explorer:



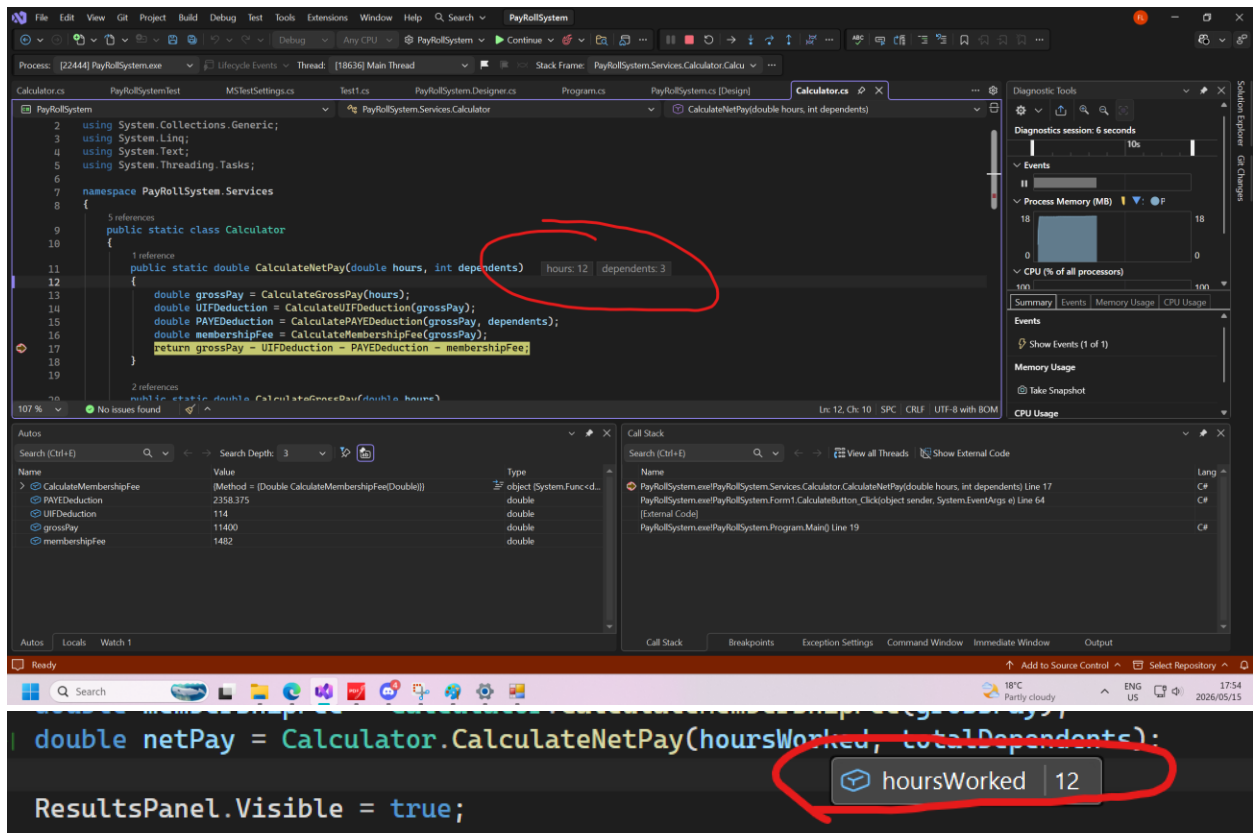
# Integration Testing

Breakpoints were used to ensure that all components of the system were interacting and working together correctly.

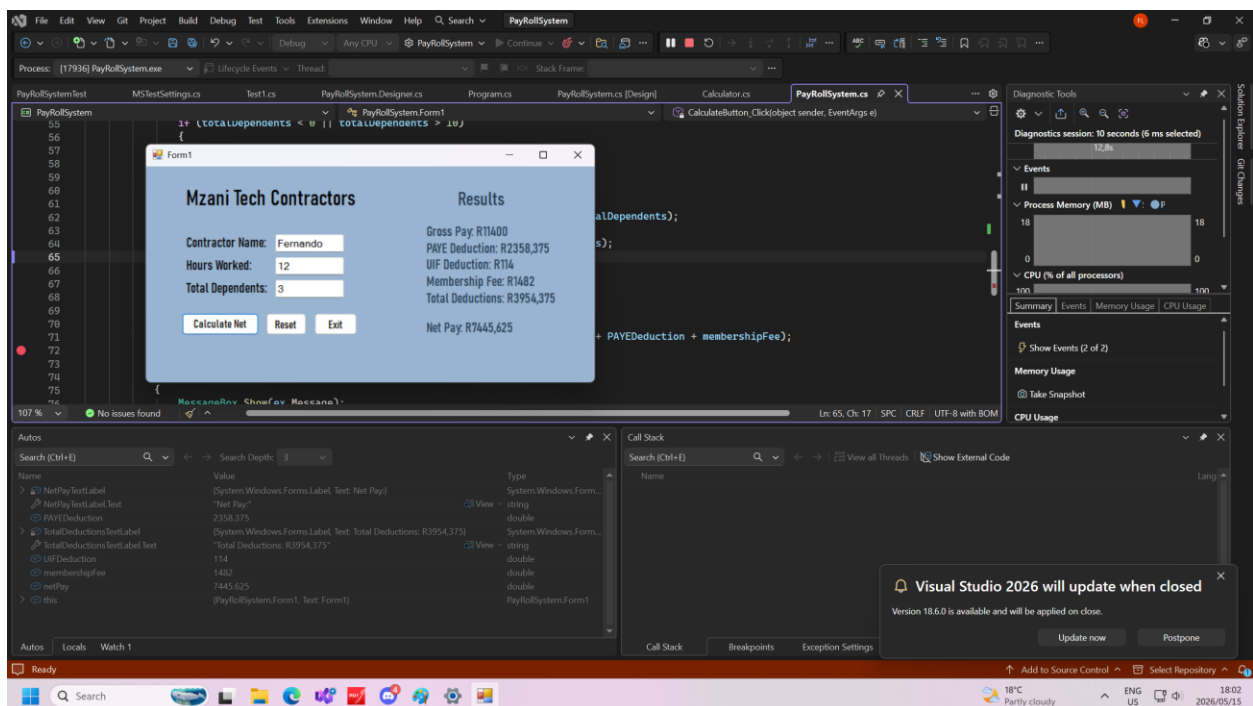
Input:



Calculation:



Output:





## System Testing

| Step | Action                                             | Expected Result                                          |
|------|----------------------------------------------------|----------------------------------------------------------|
| 1    | Pressing the 'calculate' button with valid input   | Display results                                          |
| 2    | Pressing the 'calculate' button with invalid input | Display message box with relevant error message          |
| 3    | Pressing the 'calculate' button with no input      | Display message box with relevant error message          |
| 4    | Pressing the 'clear' button                        | Empties the input fields and hides the 'results' section |
| 5    | Pressing the 'exit' button                         | Exits the application                                    |

## Retesting

After fixing a bug related to the calculation of the PAYE deduction, the system was retested. After retesting the system, the system was confirmed to be fully functional and without defects.



## Regression Testing

| Action                                             | Expected Result                                          | Actual Result                                            | Defects Found        | Corrections Made                                       |
|----------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------------------|--------------------------------------------------------|
| Pressing the 'calculate' button with valid input   | Display Results                                          | Display inaccurate results                               | Typo in calculations | Adjusted the calculations to use double instead of int |
| Pressing the 'calculate' button with invalid input | Display message box with relevant error message          | Display message box with relevant error message          | None                 | N/A                                                    |
| Pressing the 'calculate' button with no input      | Display message box with relevant error message          | Display message box with relevant error message          | None                 | N/A                                                    |
| Pressing the 'clear' button                        | Empties the input fields and hides the 'results' section | Empties the input fields and hides the 'results' section | None                 | N/A                                                    |
| Pressing the 'exit' button                         | Exits the application                                    | Exits the application                                    | None                 | N/A                                                    |

After all testing and troubleshooting steps, the system is free of issues and remains in a usable state.